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SUBJECT CODE NO: - RNP_8026
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A) (NEP) Sem - II
Examination November / December 2025 (To be held in February-2026)
Basics of Mechanical

[Time: 3 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

1. Use of non-programmable calculator is allowed
2. Q.no.1 is compulsory
3. Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

PART A

Q1 Attempt any 10 question from following

20

1. What is the primary objective of thermodynamics?
 - a. To study heat energy conversion into work
 - b. To study energy transfer and its conversion into work
 - c. To study energy conservation in engines
 - d. To study the behavior of materials under temperature variation
2. Which of the following is NOT a thermodynamic property?
 - a. Pressure
 - b. Temperature
 - c. Volume
 - d. Velocity
3. Which type of energy transfer is responsible for the transfer of heat in liquids and gases?
 - a. Conduction
 - b. Convection
 - c. Radiation
 - d. Absorption
4. What does the First Law of Thermodynamics state?
 - a. Energy can neither be created nor destroyed
 - b. Heat can never be completely converted into work
 - c. Work done on the system is equal to heat supplied to the system
 - d. Heat flows naturally from a hot body to a cold body
5. Which of the following is a major component of a boiler?
 - a. Heat exchanger
 - b. Pump
 - c. Furnace
 - d. Turbine
6. The primary function of the boiler mountings is
 - a. To support the boiler structure
 - b. To regulate pressure and water level

- c. To increase the efficiency
 - d. To reduce heat loss
7. The main function of a pulley in power transmission is to
- a. Reduce energy consumption
 - b. Change direction of motion
 - c. Increase torque
 - d. Store energy
8. Which of the following machine tools uses rotary cutters to remove material?
- a. Drilling machine
 - b. Lathe machine
 - c. Milling machine
 - d. Grinding machine
9. Which of the following heat treatment processes is used to reduce brittleness in steel?
- a. Annealing
 - b. Normalizing
 - c. Hardening
 - d. Tempering
10. Which of the following processes involves heating the material to a specific temperature and then cooling it rapidly to increase hardness?
- a. Normalizing
 - b. Annealing
 - c. Hardening
 - d. Tempering
11. Which of the following machine tools is primarily used for cutting and shaping metal components by rotating a workpiece against a cutting tool?
- a. Lathe Machine
 - b. Milling Machine
 - c. Drilling Machine
 - d. Shaper Machine
12. In a milling machine, the cutting tool rotates while the workpiece is moved in which direction?
- a. Vertically
 - b. Horizontally
 - c. Both vertically and horizontally
 - d. Diagonally

PART B

- Q.2 a) Explain thermodynamics equilibrium in details. 5
 b) What is pressure? Explain its any one type of pressure measuring device. 5
- Q.3 a) State the various modes of heat transfer and explain them briefly. 5
 b) Write short note on forms of energy. 5
- Q.4 a) Explain working principle of Refrigerator. 5

- b) Explain 4 stroke SI engine. **5**
- Q.5 a) State and explain the selection criteria for Engineering Materials. **5**
b) What is heat treatment process? Explain in detail normalizing heat treatment process. **5**
- Q.6 a) Explain with neat sketch the various operations carried out on drilling machine. **5**
b) Explain CNC in Details. **5**
- Q.7 a) Differentiate between flat and v belt drive. **5**
b) Explain the construction and working of single plate clutch with neat diagram. **5**